सहायक प्राध्यापक (भूगर्भशास्त्र) महाराष्ट्र शिक्षण सेवा, गट-अ
(महाविद्यालयीन शाखा)
उच्च व तंत्र शिक्षण विभाग
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Higher and Technical Education Department

परीक्षेचे टप्पे: **१)** लेखी परीक्षा - २०० गुण **२)** मुलाखत- ५० गुण

-: परीक्षा योजना :-

विषय व सांकेतांक	माध्यम	प्रश्नसंख्या	गुण	कालावधी	दर्जा	प्रश्नपत्रिकेचे स्वरुप
विषयांशी संबंधीत घटक (११५२)	इंग्रजी	१००	२००	एक तास	पदव्युत्तर पदवी	वस्तुनिष्ठ बहुपर्यायी

अ) नकारात्मक गुणदान -

१) प्रत्येक चुकीच्या उत्तराकरीता २५% किंवा १/४ एवढे गुण एकूण गुणांमधून वजा/ कमी करण्यात येतील.
२) एखाद्या प्रश्नाची एकापेक्षा अधिक उत्तरे दिली असल्यास अथवा ज्या उमेदवाराने उत्तरपत्रिकेत पूर्ण वर्तुळ चिन्हांकित केले नसेल अशा प्रश्नाचे उत्तर चुकीचे समजण्यात येऊन त्या प्रश्नाच्या उत्तराकरीता २५% किंवा १/४ एवढे गुण एकूण गुणांमधून वजा/कमी करण्यात येतील.
३) वरीलप्रमाणे कार्यपध्दतीचा अवलंब करताना एकूण अंतिम गुणांची बेरीज अपूर्णांकात आली तरीही ती अपूर्णांकातच राहील व पुढील कार्यवाही त्याच्या आधारे करण्यात येईल.
४) एखाद्या प्रश्नाचे उत्तर अनुत्तरित असेल तर, अशा प्रकरणी नकारात्मक गुणांची पध्दत लागू असणार नाही.

ब) अंतिम गुणवत्ता यादी ही वस्तुनिष्ठ परीक्षेतील व मुलाखतीतील एकत्रित गुणांवर आधारीत राहील.

-: अभ्यासक्रम :-

विषयांशी संबंधित घटक यामध्ये खालील घटक व उपघटकांचा समावेश असेल.

अ. क्र.	घटक व उपघटक
1.	**The Earth and the Solar System** Milky Way Galaxy and the Solar system;Kepler's laws of planetary motion;Modern theories on the origin of the Earth and other planetary bodies;Meteorites;Earth's orbital parameters, Geological Time Scale; Chemical composition and the primary differentiation of the earth., Earth's gravity and magnetic fields and its thermal structure; Isostasy; Radioactive isotopes and their applications.
2.	**Earth Materials, Surface Features and Processes** Gross composition and physical properties of important minerals and rocks; properties and processes responsible for mineral concentrations; nature and distribution of rocks and minerals in different units of the earth and different parts of India; Physiography of the Earth; weathering, erosion, transportation and deposition of Earth's material; formation of soil, sediments and sedimentary rocks; energy balance of the Earth's surface processes; physiographic features and river basins in India.
3.	**Interior of the Earth, Deformation and Tectonics** Basic concepts of seismology and internal structure of the Earth; Physico-chemical and seismic properties of Earth's interior; Concepts of stress and strain; Behaviour of rocks under stress; Folds, faults and joints; Earthquakes – their causes and measurement; Interplate and intraplate seismicity; Paleomagnetism, sea floor spreading and plate tectonics.
4.	**Oceans and Atmosphere** Hypsography of the continents and ocean floor; continental shelf, slope, rise and abyssal plains; Physical and chemical properties of seawater and their spatial variations; Residence times of elements in seawater; Ocean currents, waves and tides, important current systems, thermohaline circulation and the oceanic conveyor belt; Major water masses of the world's oceans; Biological productivity in the oceans. Motion of fluids, waves in atmospheric and oceanic systems; Atmospheric turbulence and boundary layer; Structure and chemical composition of the atmosphere, lapse rate and stability, scale height,

	geopotential, greenhouse gases and global warming; Cloud formation and precipitation processes, air-sea interactions on different space and time scales; Insolation and heat budget, radiation balance, general circulation of the atmosphere and ocean; Climatic and sea level changes on different time scales; Coupled ocean-atmosphere system, Indian ocean dipole; El Nino Southern Oscillation (ENSO); General weather systems of India, Monsoon system, cyclones and jet stream, Western disturbances and severe local convective systems, distribution of precipitation over India; Marine and atmospheric pollution, ozone depletion.
5.	**Environmental Earth Science** Properties of water; hydrological cycle; water resources and management; Energy resources, uses, degradation, alternatives and management; Ecology and biodiversity; Impact of use of energy and land on the environment; Exploitation and conservation of mineral and other natural resources; Natural hazards.
6.	**Mineralogy and Petrology** Concept of point group, space group, reciprocal lattice; Diffraction and imaging; Concepts of crystal field theory and mineralogical spectroscopy; Lattice defects (point, line and planar); Electrical, magnetic and optical properties of minerals; Bonding and crystal structures of common oxides, sulphides, and silicates; Transformation of minerals – polymorphism, polytypism, and polysomatism; Solid solution and exsolution. Steady-state geotherms; Genesis, properties, emplacement and crystallization of magmas; Phase equilibrium studies of simple systems, effect of volatiles on melt equilibria; Magma- mixing, mingling and immiscibility; Classification of Igneous rocks, historic perspective and the IUGS systematic Petrogenetic aspects of important rock suites of India, such as the Deccan Traps, layered igneous complexes, anorthosites, carbonatites, alkaline rocks, kimberlites, ophiolites and granitoids. Metamorphic structures and textures; isograds and facies; Mineral reactions with condensed phases, solid solutions, mixed volatile equilibria and thermo-barometry; Metamorphism of pelites, mafic-ultra mafic rocks; Material transport during metamorphism; P-T-t path in regional metamorphic terrains;Plate tectonics and metamorphism; Important metamorphic rock suites of India such as Charnockites, Khondalites, Marbles and Granulites.
7.	**Structural Geology and Geotectonics** Theory of stress and strain;behaviour of rocks under stress;various states of stress and their representation by Mohrcircles; Application of Mohr Circle methods; different types of failure and sliding criteria; Geometry and mechanics of fracturing and conditions for reactivation of pre-existing discontinuities; common types of finite strain ellipsoids; L-S tectonic fabrics. Techniques of strain analysis;particle paths and flow patterns;progressive strain history; Introduction to deformation mechanisms; Role of fluids in deformation processes; Shear Zone analysis; geometry and analyses of brittle-ductile and ductile shear zones; geometry and mechanics of development of folds, foliations and lineations; Interference patterns of superposed fold; Fault-related folding; Gravity induced structures; Tectonic features of extensional-,compressional-,and strike-slip-terrains and relevance to plate boundaries; mantle plumes.
8.	**Plate Tectonics and Geodynamics** Marine magnetic anomalies, sea floor spreading; mid-oceanic ridges and geodynamics; plate tectonics hypothesis; plate boundaries and seismicity; Heat flow mechanisms, thermal modeling of earth, core-mantle convection and mantle plumes; Himalayan Orogeny; concept of super continent, their assembly and breakup.
9.	**Paleontology and its applications** Theories on the origin of life; Organic evolution – Punctuated Equilibrium and Phyletic Gradualism models; Mass extinctions and their causes; Application of fossils in age determination and correlation; Paleoecology, Life habitats and various ecosystems, Paleobiogeography; modes of preservation of fossils and taphonomic considerations; types of microfossils; Environmental significance of fossils and trace fossils; use of microfossils in interpretation of sea floor tectonism; application of micropaleontology in hydrocarbon exploration; Oxygen and Carbon isotope studies of microfossils and their use in paleo-oceanographic and paleoclimatic interpretation; Important invertebrate fossils, vertebrate fossils, plant fossils and microfossils in Indian stratigraphy.
10.	**Sedimentology and Stratigraphy** Classification of sediments and sedimentary rocks; clastic, volcanoclastic and chemical sedimentation; flow regimes and processes of sediment transport. Sedimentary textures and structures; sedimentary facies and environments, reconstruction of paleoenvironments; formation and evolution of sedimentary basins; diagenesis of siliciclastic and

	carbonate rocks. Recent developments in stratigraphic classification; Code of stratigraphic nomenclature – Stratotypes, Global Boundary Stratotype Sections and Points (GSSP); Lithostratigraphic, chronostratigraphic and biostratigraphic subdivisions; methods of startigraphic correlation including Shaw's Graphic correlation; concept of sequence stratigraphy; rates of sediment accumulation, unconformities; Facies concept in Stratigraphy–Walther's law; methods for paleogeographic reconstruction; Earth's Climatic History; Phanerozoic stratigraphy of India with reference to the type areas– their correlation with equivalent formations in other regions; Boundary problems in Indian Phanerozoic stratigraphy.
11.	**Marine Geology and Paleoceanography** Morphologic and tectonic domains of the ocean floor; Structure, composition and mechanism of the formation of oceanic crust; hydrothermal vents; Ocean margins and their significance; Ocean Circulation, Coriolis effect and Ekman spiral, convergence, divergence and upwelling; thermohaline circulation and oceanic conveyor belt;formation of bottom waters; major water masses of the world's oceans; factors controlling the deposition and distribution of oceanic sediments; geochronology of oceanic sediments, diagenetic changes in oxic and anoxic environments; Tectonic evolution of the ocean basins; Mineral resources. Paleoceanography – Approaches to paleoceanographic reconstructions; various proxy indicators for paleoceanographic interpretation; reconstruction of monsoon variability by using marine proxy records; Opening and closing of ocean gateways and their effect on circulation and climate during the Cenozoic; Sea level processes and sea level changes; Sea surface temperatures.
12.	**Geochemistry** Atomic structure and properties of elements, Periodic Table; Ionic substitution in minerals; Phase rule and its applications in petrology, thermodynamics of reactions involving pure phases, ideal and non-ideal solutions and fluids; Equilibrium and distribution coefficients; nucleation and diffusion processes in igneous, metamorphic and sedimentary environments, redox reactions and Eh- pH diagrams and their applications; Mineral assemblages as 'sensors' of ambient environments; geochemical studies of aerosols, surface-and marine waters; Radioactive decay schemes and their application to geochronology and petrogenesis; Stable isotopes and their application to earth system processes; geochemical differentiation of the earth, geochemical cycles.
13.	**Economic Geology** Magmatic, hydrothermal and surface processes of ore formation;genetic classification of ore deposits; Metallogeny and its relation to crustal evolution; ore-forming systems; methods of mineral deposit studies including ore microscopy, fluid inclusions and isotopic systematics; Ores and metamorphism- cause and effect relationships; Geological setting, characteristics, and genesis of ferrous, base and noble metals; detailed study of major Indian mineral deposits; National Mineral Policy and Mineral Economics. Origin, migration and entrapment of petroleum; Properties of source and reservoir rocks; structural, stratigraphic and combination traps; methods of petroleum exploration; Concepts of petrophysics; Petroliferous basins of India. Origin of peat, lignite, bitumen and anthracite; Classification, rank and grading of coal, coal petrography; Coal bed methane; coal resources of India; Gas hydrates; Nuclear and non-conventional energy resources.
14.	**Precambrian Geology and Crustal Evolution** Evolution of lithosphere, hydrosphere, atmosphere,biosphere,and cryosphere; Lithological, geochemical and stratigraphic characteristics of granite – greenstone and granulite belts. Stratigraphy and geochronology of the cratonic nuclei, mobile belts and Proterozoic sedimentary basins of India; Life in Precambrian; Precambrian–Cambrian boundary with special reference to India.
15.	**Quaternary Geology** Definition and subdivisions of the Quaternary period; Quaternary Stratigraphy – Marine Oxygen Isotope stratigraphy, biostratigraphy and magnetostratigraphy; Quaternary climates – glacial-interglacial cycles, eustatic changes; Proxy indicators of paleoenvironmental/ paleoclimatic changes, Quaternary records from land, ocean and cryosphere (ice core studies); Responses of geomorphic systems to climate; Sea level and tectonics on variable time scales; Quaternary dating methods, – radiocarbon, uranium series, luminescence; Quaternary stratigraphy of India– continental records (fluvial, glacial, aeolian, palaeosols and duricrust), marine records; Evolution of man and Stone Age cultures; Plant and animal life in relation to glacial and interglacial cycles during Quaternary.
16.	**Climatology** Fundamental principles of climatology; Earth's radiation balance; latitudinal and seasonal variation

	of insolation, temperature, pressure, wind belts, humidity; Cloud formation and precipitation; Water balance; Air masses, monsoon, Jet streams, tropical cyclones and ENSO. Classification of climates – Koppen's and Thornthwaite's scheme of classification; Climate change.
17.	**Geomorphology** Concepts in geomorphology; historical and process Geomorphology; landforms in relation to climate, rock type, structure and tectonics; Processes – weathering, pedogenesis, mass movement, erosion, transportation and deposition; Geomorphic processes and landforms – fluvial, glacial, eolian, coastal and karst; River forms and processes – stream flow, stage- discharge relationship; hydrographs and flood frequency analysis; Submarine relief; geomorphology and topographic analysis including DEM; environmental change– causes, effects on processes and landforms; Extra-terrestrial geomorphology.
18.	**Remote Sensing, GIS and GPS** Elements of photo-interpretation; Electromagnetic spectrum, emission range energy sources, energy interaction in the atmosphere, atmospheric windows, atmospheric effects on remotely sensed data, sensors; Geological interpretations of aerial photos and imageries, image enhancement, digital analysis; Microwave remote sensing for geological applications; Indian Remote Sensing satellite programmes. GIS - Preparation of raster and vector database and maps; Data structure, attribute data, design, query, and analysis; Data models, spatial data structures and spatial operations; DEM, interpolation techniques; GIS tools and applications. Global positioning systems - GPS data collection and processing techniques; Differential GPS (DGPS) and real-time kinematic (RTK) positioning; Geodetic Datum and Coordinate Systems; Recent advancements in GPS technology and applications in geosciences.
19.	**Satellite Meteorology** Meteorological satellites; Types - Polar-orbiting and geostationary satellites; Payloads - Visible and infrared radiometers, microwave sensors, multi-scanner radiometers; Remote Sensing Applications; Identification of synoptic systems, fog, sandstorms, and cyclones; Sea Surface Temperature (SST) estimation, cloud top temperature retrieval, wind and rainfall measurements; Atmospheric temperature and humidity soundings; Microwave and hyperspectral observations for weather and climate studies; Satellite-based precipitation estimation (e.g., TRMM, GPM); Data assimilation in numerical weather prediction (NWP); Detection of extreme weather events and long-term climate trends; Aerosol and greenhouse gas monitoring.
20.	**Engineering Geology** Engineering properties of rocks and physical characteristics of building stones, concretes and other aggregates; geological investigations for construction of dams, bridges, highways and tunnels; mass movements with special emphasis on landslides and solutions for hillslope stability; geo-material in engineering construction; types of synthetic materials used as remedial measures; seismic design of buildings.
21.	**Hydrogeology** Different components of the hydrological cycle; Unit hydrograph; Characteristics of unsaturated and saturated zones; Vertical distribution of subsurface water; Darcy's law, limitations and validity; Hydrogeological classification of water-bearing formations; types of aquifers in different geological settings; principles of groundwater flow; steady and unsteady state; Aquifer Performance Test; Tracing of groundwater movement with flow net; measurement of groundwater recharge, various artificial recharge techniques, rainwater harvesting; Managed Aquifer Recharge; Hydrogeochemistry - geochemical processes in the groundwater, application of stable isotopes;quality of groundwater and suitability for various purposes;fresh and salt-water relationships in coastal areas; Groundwater provinces in India.
22.	**Exploration Geology** Geological, geophysical, geochemical and geobotanical methods of surface and sub- surface exploration on different scales; Sampling, assaying and evaluation of mineral deposits.
23.	**Gravity and Magnetic fields of the earth** Normal gravity field; Clairaut's theorem; Shape of the earth,geoid; Free-air, Bouguer and isostatic anomalies, isostatic models for local and regional compensation; Geomagnetic field, secular and transient variations and their theories; Palaeomagnetism, construction of polar wandering curves.
24.	**Gravity and Magnetic Methods** Gravimeters and magnetometers; data acquisition from land, air and under water; corrections and reduction of anomalies, ambiguity, interpretation of anomalies of simple geometric bodies; Single pole, sphere, horizontal cylinder, sheet, dyke and fault; Forward modelling, inversion of arbitrary

	shaped bodies and 2-D, 3-D interfaces.
25.	**Seismology** Seismometry - short period, long period, broad band and strong motion; Elements of earthquake seismology; seismic sources –faulting source, double couple hypothesis, seismic moment tensor; Focal mechanism and fault plane solutions; Seismic gaps; Seismotectonics and structure of the earth; Himalayan and stable continental region earthquakes; Reservoir induced seismicity; seismic hazards; earthquake prediction, travel time residuals, velocity anomalies; Seismic tomography.
26.	**Seismic Methods** Snell's Law; Ray theory; Reflection, refraction, diffraction; Seismic energy sources; Detectors, seismic data recording; Seismic noises and noise profile analysis; reduction to a datum and weathering corrections; Interpretation of refraction and reflection data; Time-distance graph, Horizontal and dipping interfaces, CDP/CMP, velocity analysis, F-K filtering, stacking, deconvolution, migration before and after stack; bright spot analysis; wavelet processing; attenuation studies, MASW, shear waves, AVO; VSP; introduction to 3D seismic imaging; Seismic stratigraphy.
27.	**Electrical and Electromagnetic Methods** Principles of electrical method – electrical properties of rocks; electrical resistivity method; electrode array configurations, profiling and sounding; resistivity transform and direct interpretation; curve matching techniques; electrical resistivity imaging; Self-potential method; Induced polarization method. Electromagnetic field techniques; elliptic polarization, in-phase and out of phase components, horizontal and vertical loop methods; Data interpretation; VLF (very low frequency); AFMAG (Audio frequency magnetic) methods; Central frequency sounding; Transient electromagnetic methods; Magneto-telluric method; GPR.
28.	**Well logging** Open hole, cased hole and production logging; electrical logs; lateral, induction, temperature, S.P; porosity logs; sonic, density, neutron; natural gamma; determination of formation factor, water saturation, lithology; logging while drilling.
29.	**Environmental Geography** Man-land relationship; Resources – renewable and non-renewable; Natural and man-made hazards – droughts, floods, cyclones, earthquakes, landslides, tsunamis; ecological balance; environmental pollution and deterioration.
30.	**Geography of India** Physiography, drainage, climate, soils and natural resources – the Himalaya, Ganga- Brahmaputra Plains, and Peninsular India Precambrian shield, the Gondwana rift, Deccan Plateau; Indian climatology with special reference to seasonal distribution and variation of temperature, humidity, wind and precipitation; Climate zones of India; Agricultural geography of India; Population – its distribution and characteristics; urbanization and migration; Environmental problems and issues.
31.	**Physical Oceanography** Water masses and mixing - T-S diagrams; Mixing processes in the oceans; characteristics of major water masses; waves and coastal processes - Wind generated waves, shallow and deep-water waves, propagation,principles of wave forecasting. Tides - tide producing forces, harmonic method; Tidal currents in shallow seas, estuaries and rivers; Coastal processes – wave transformation stratification effects, bottom friction, wave reflection, refraction and diffraction; Breakers and surf; Littoral currents; Wave action on sediments – movement of beach material; Rip currents; Beach stability; Ocean beach nourishment; Harbour resonance; Seiches; Tsunamis. Estuarine and coastal dynamics; Estuaries - classification and nomenclature; Circulation, mixing models, sedimentation, salinity intrusion, effect of stratification; coastal pollution - pollutant dispersion in estuaries and near-shore areas; coastal zone management. Ocean circulation and Large-scale Motion; wind-driven ocean motion; Ekman's theory; Sverdrup-Stommel- Munk theories; upwelling and sinking (with reference to the Indian ocean); divergences, convergences, barotropic/baroclinic conditions, oceanic eddies. Global conveyor belt circulation, subtropical gyres, western boundary currents, equatorial currents; El Niño, monsoonal winds and currents over the North Indian Ocean, Somali current; Southern Ocean; Arabian Sea upwelling.
32.	**Chemical Oceanography:** Seawater composition and element distribution; Major and minor elements, classification by distribution, chemical exchanges across interfaces, residence times; Element chemistry in specialized environments; Hydrothermal vents; Anoxic basins; Sediment pore fluids; Anthropogenic influences.

	Chemical and biological interactions; Ionic interactions; Biogeochemical cycling of nutrients, trace metals, and organic matter;Air-sea exchange of biogenic dissolved gases; CO_2-carbonate system, alkalinity, pH regulation, biological pump. Factors influencing $CaCO_3$, silicates, manganese nodules, phosphorites, and massive sulfide deposits.

दिनांक – २१ जुलै, २०२५

अवर सचिव
महाराष्ट्र लोकसेवा आयोग